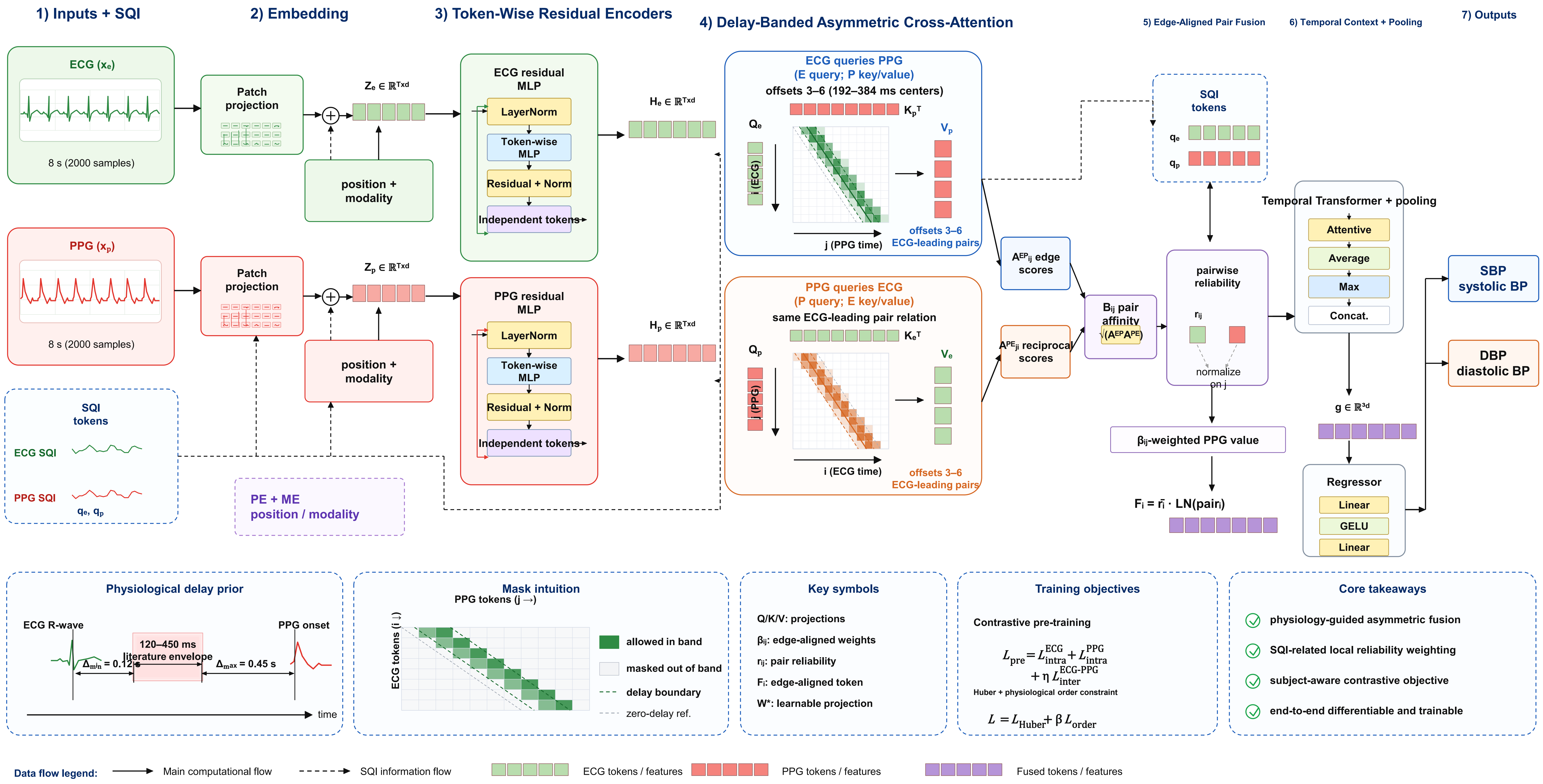
1) Inputs + SQI
2) Embedding
3) Token-Wise Residual Encoders
4) Delay-Banded Asymmetric Cross-Attention
5) Edge-Aligned Pair Fusion
6) Temporal Context + Pooling
7) Outputs
ECG (x_e)
8 s (2000 samples)
PPG (x_p)
8 s (2000 samples)
Patch projection
Patch projection
$Z_e \in \mathbb{R}^{T\times d}$
$Z_p \in \mathbb{R}^{T\times d}$
position + modality
position + modality
ECG residual MLP
PPG residual MLP
LayerNorm
Token-wise MLP
Residual + Norm
Independent tokens
$H_e \in \mathbb{R}^{T\times d}$
$H_p \in \mathbb{R}^{T\times d}$
SQI tokens
ECG SQI
PPG SQI
q_e, q_p
PE + ME
position / modality
ECG queries PPG
(E query; P key/value)
offsets 3–6 (192–384 ms centers)
Q_e
K_p^T
V_p
i (ECG)
j (PPG time)
offsets 3–6
ECG-leading pairs
PPG queries ECG
(P query; E key/value)
same ECG-leading pair relation
Q_p
K_e^T
V_e
j (PPG)
i (ECG time)
offsets 3–6
ECG-leading pairs
A^{EP}_{ij} edge scores
A^{PE}_{ji} reciprocal scores
B_{ij} pair affinity
$\sqrt{(A^{EP}A^{PE})}$
SQI tokens
q_e
q_p
pairwise reliability
r_{ij}
normalize on j
β_{ij}-weighted PPG value
$F_i = \bar{r}_i \cdot \mathrm{LN}(\mathrm{pair}_i)$
Temporal Transformer + pooling
Attentive
Average
Max
Concat.
$g \in \mathbb{R}^{3d}$
Regressor
Linear
GELU
Linear
SBP systolic BP
DBP diastolic BP
Physiological delay prior
ECG R-wave
PPG onset
120–450 ms
literature envelope
Δ_{min} = 0.12 s
Δ_{max} = 0.45 s
time
Mask intuition
PPG tokens (j →)
ECG tokens (i ↓)
allowed in band
masked out of band
delay boundary
zero-delay ref.
Key symbols
Q/K/V: projections
β_{ij}: edge-aligned weights
r_{ij}: pair reliability
F_i: edge-aligned token
W*: learnable projection
Training objectives
Contrastive pre-training
$L_{pre} = L^{ECG}_{intra} + L^{PPG}_{intra} + \eta\, L^{ECG\text{-}PPG}_{inter}$
Huber + physiological order constraint
$L = L_{Huber} + \beta\, L_{order}$
Core takeaways
physiology-guided asymmetric fusion
SQI-related local reliability weighting
subject-aware contrastive objective
end-to-end differentiable and trainable
Data flow legend:
Main computational flow
SQI information flow
ECG tokens / features
PPG tokens / features
Fused tokens / features